

An Explainer on Gauge Fields

From connections to the modern point of view

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These notes are an attempt to fully decode statements like “ A is a $U(1)$ -gauge field,” “ μ is a $\mathbb{Z}_2$ -gauge field,” and “ a is a background gauge field for the global symmetry G .” We start from the physics you already know (Maxwell, the covariant derivative, Aharonov–Bohm), build up to the geometric definition (a gauge field is a *connection* on a *principal bundle*), make sense of *finite*-group gauge fields (which have no Lie algebra), and then turn to the distinction between **background** and **dynamical** gauge fields. That distinction is the doorway to the modern, “Seiberg-style” point of view, in which a quantum field theory is understood through how its symmetries couple to background fields, what happens when you **gauge** them, and what obstructs gauging (**t Hooft anomalies**). We close with a working preview of generalized (higher-form) global symmetries.

Prerequisites. A first course in QFT (path integrals, Noether’s theorem), basic Lie groups/algebras, and enough differential geometry to be unafraid of differential forms. No bundle theory is assumed; we build what we need.

How to read. Boxes labelled Definition/Theorem are the load-bearing statements; Intuition and Note boxes are the “how a practitioner thinks” commentary. Green Exercise boxes are scattered throughout and tagged (**easy**), (**classic**), (**medium**), or (**thinker**)—do them as you go. End-of-section payoffs are summarized in yellow Equation boxes. A reference list and a “books to buy” list are at the end.

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1 Big picture: what problem are gauge fields solving?

Let me start with the punchline and the motivation, because if you only remember one paragraph, it should be this one.

Idea 1.1 (The one-paragraph summary)

A **gauge field** is a *connection*: a rule for comparing internal data (the phase of a wavefunction, the color of a quark, an order parameter) at different points of spacetime. “ G -gauge field” means the comparison is done using the group G . When G is continuous (like $U(1)$ or $SU(3)$) the connection is the familiar potential A_μ ; when G is finite (like $\mathbb{Z}_2$) the connection is a discrete assignment of group elements to paths. A **dynamical** gauge field is summed over in the path integral—it is part of the theory’s degrees of freedom. A **background** gauge field is fixed by hand—it is a knob you turn to probe a *global* symmetry. The single most important modern idea is that you learn almost everything about a symmetry by coupling it to a background gauge field and asking how the partition function $Z[A]$ responds; **gauging** the symmetry then means promoting that background to a dynamical field (summing over it).

Everything below is an unpacking of this paragraph.

1.1 Three things you already know that are secretly the same thing

You have already met connections at least three times. Recognizing them as the same object is most of the battle.

Example 1.1 (Electromagnetism: the vector potential)

In E&M you write $\mathbf{E}$ and $\mathbf{B}$ in terms of a potential, $\mathbf{B} = \nabla \times \mathbf{A}$, $\mathbf{E} = -\nabla\phi - \partial_t \mathbf{A}$. The potential $A_\mu = (\phi, \mathbf{A})$ is not unique: $A_\mu \rightarrow A_\mu + \partial_\mu \lambda$ leaves $\mathbf{E}, \mathbf{B}$ unchanged. That redundancy is **gauge invariance**, and A_μ is a $U(1)$ -gauge field. The physically detectable content of A_μ is not its value at a point but its *holonomy* $\oint A$ around loops—this is exactly the Aharonov–Bohm phase.

Example 1.2 (General relativity: the Christoffel connection)

To differentiate a vector field on a curved manifold you cannot just subtract vectors at different points—they live in different tangent spaces. You need a rule to *transport* one to the other: the connection $\Gamma_{\mu\nu}^\rho$, packaged into the covariant derivative $\nabla_\mu V^\rho = \partial_\mu V^\rho + \Gamma_{\mu\nu}^\rho V^\nu$. Its curvature is the Riemann tensor. A Yang–Mills gauge field is the *same kind of object*: a connection, but on an “internal” space (a charge phase, a color index) rather than the tangent space. The dictionary $\Gamma \leftrightarrow A$, Riemann $\leftrightarrow F$ is exact, and worth keeping in your head throughout.

Example 1.3 (Quantum mechanics: the Berry connection)

Adiabatically drag a quantum system around a loop in parameter space and it picks up a geometric phase $\exp(i \oint \mathcal{A})$, the Berry phase, where $\mathcal{A} = i\langle n | dn \rangle$ is the Berry connection. Again: a connection, whose gauge-invariant content is a holonomy. The phase ambiguity of the

instantaneous eigenstate $|n\rangle \rightarrow e^{i\theta}|n\rangle$ is a $U(1)$ gauge transformation.

Intuition 1.1 (The common thread)

Each example has (i) some internal data attached to each spacetime/parameter point (a phase, a tangent vector, a state), (ii) an ambiguity in how we label that data (the gauge freedom), and (iii) a *connection* that tells us how to compare data at neighboring points despite the ambiguity. The connection is the gauge field. Its curvature measures the failure of comparison-around-a-small-loop to be trivial; its holonomy is the physical, gauge-invariant output.

1.2 Why a hep-th student should care

- **The Standard Model *is* a gauge theory.** It is the statement that nature has a dynamical $SU(3) \times SU(2) \times U(1)$ connection. Every force except gravity is the curvature of some connection; gravity is the curvature of the spacetime connection. “Understanding gauge fields” is understanding the grammar of fundamental physics.
- **Topology enters physics through gauge fields.** Monopole charge, instanton number, θ -angles, the quantization of flux, anomalies—these are all *topological* invariants of gauge field configurations. This is where QFT and geometry/topology genuinely merge.
- **The modern reorganization of QFT is symmetry-first.** Since Gaiotto–Kapustin–Seiberg–Willett (2014), the community thinks of a QFT through the lens of its (generalized) symmetries, realized as topological defect operators and probed by background gauge fields. Anomalies and dualities become statements about these backgrounds. If you want to read current hep-th, you must be fluent in “coupling to a background gauge field,” “gauging,” and “’t Hooft anomaly.”
- **Condensed matter speaks the same language.** Topological order, SPT phases, the toric code, fractionalization—all are phrased in terms of (often *finite*) gauge fields and their anomalies. This is why $\mathbb{Z}_2$ gauge fields are worth as much of your attention as $U(1)$ ones.

Note 1.1 (A map of these notes). **Part I** (§2–§4): ordinary continuous gauge fields, ending at the geometric definition (connections on principal bundles). **Part II** (§5): finite-group gauge fields, where the lattice/cochain picture is unavoidable and clarifying. **Part III** (§6–§9): background vs. dynamical, gauging, ’t Hooft anomalies, and the modern point of view with a preview of higher-form symmetries.

2 From global to local: gauging a symmetry, the physicist's way

We begin with the move that historically *produced* gauge fields, and that is still the fastest route to the covariant derivative: take a global symmetry and demand that it hold locally.

2.1 A free complex scalar and its global U(1)

Consider a complex scalar field $\phi(x)$ with action

$$S[\phi] = \int d^d x \left(\partial_\mu \phi^* \partial^\mu \phi - m^2 \phi^* \phi \right).$$

This has a **global** U(1) symmetry: for a *constant* $\alpha \in \mathbb{R}$,

$$\phi(x) \longmapsto e^{i\alpha} \phi(x).$$

“Global” means α is the same at every spacetime point. The kinetic term is invariant because $\partial_\mu(e^{i\alpha}\phi) = e^{i\alpha}\partial_\mu\phi$ —the constant phase slides through the derivative.

Now ask: what if we want the phase rotation to be a symmetry even when $\alpha = \alpha(x)$ varies from point to point? This is the demand of **local** (gauge) symmetry. The mass term is still fine, but the derivative is not:

$$\partial_\mu(e^{i\alpha(x)}\phi) = e^{i\alpha(x)}(\partial_\mu\phi + i(\partial_\mu\alpha)\phi).$$

The offending term $i(\partial_\mu\alpha)\phi$ appears because the derivative compares $\phi(x+dx)$ with $\phi(x)$, but these are now rotated by *different* phases. The derivative is comparing apples to oranges.

Intuition 2.1 (Why the naive derivative fails)

$\partial_\mu\phi \approx (\phi(x+\epsilon\hat{\mu}) - \phi(x))/\epsilon$ subtracts the field at two different points. Under a local transformation those two points get rotated by $e^{i\alpha(x+\epsilon)}$ and $e^{i\alpha(x)}$ respectively. To subtract meaningfully we must first *transport* $\phi(x+\epsilon)$ back to x , undoing the relative rotation. The object that performs this transport is the gauge field.

2.2 Introducing the connection: the covariant derivative

Introduce a new field $A_\mu(x)$ and define the **covariant derivative**

$$D_\mu\phi \stackrel{\text{def}}{=} (\partial_\mu - iA_\mu)\phi$$

(taking ϕ to have unit charge; a charge- q field uses $\partial_\mu - iqA_\mu$). We *demand* that $D_\mu\phi$ transform the same way as ϕ itself, $D_\mu\phi \rightarrow e^{i\alpha(x)}D_\mu\phi$, so that $|D_\mu\phi|^2$ is gauge invariant. Working out what A_μ must do:

$$D_\mu\phi \rightarrow (\partial_\mu - iA'_\mu)(e^{i\alpha}\phi) = e^{i\alpha}(\partial_\mu\phi + i(\partial_\mu\alpha)\phi - iA'_\mu\phi) \stackrel{!}{=} e^{i\alpha}(\partial_\mu - iA_\mu)\phi,$$

which forces

$$A_\mu \longmapsto A_\mu + \partial_\mu\alpha$$

This is exactly the gauge transformation of the electromagnetic potential. The gauge field A_μ was invented precisely to absorb the inhomogeneous $\partial_\mu\alpha$ term, restoring local symmetry.

Equation 2.1 (The gauging dictionary, abelian version)

$$\phi \rightarrow e^{i\alpha(x)}\phi, \quad D_\mu = \partial_\mu - iA_\mu, \quad A_\mu \rightarrow A_\mu + \partial_\mu\alpha, \quad D_\mu\phi \rightarrow e^{i\alpha(x)}D_\mu\phi.$$

Replace every ∂_μ by D_μ (“minimal coupling”) and a globally symmetric theory becomes locally symmetric.

Intuition 2.2 (A_μ as “how much to rotate as you step”)

Think of $A_\mu(x) dx^\mu$ as the infinitesimal phase you must rotate by to carry ϕ from x to $x + dx$ while “keeping it fixed” according to the connection. The covariant derivative subtracts *that* transported value, so it measures the genuine change of ϕ relative to the connection. A connection is, literally, a prescription for the words “keeping it fixed as you move.”

Exercise 2.1 ((easy) minimal coupling) — Show that $|D_\mu \phi|^2 = |\partial_\mu \phi|^2 + iA^\mu(\phi \partial_\mu \phi^* - \phi^* \partial_\mu \phi) + A_\mu A^\mu |\phi|^2$, and identify the term linear in A_μ as $A_\mu j^\mu$ with j^μ the Noether current of the global U(1). *Moral: minimal coupling couples A to the very current whose symmetry you are gauging.*

2.3 Curvature: the field strength

The gauge field A_μ has its own dynamics, governed by a gauge-invariant object built from it: the **field strength** (curvature)

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu = i[D_\mu, D_\nu].$$

The second equality is the cleanest definition: *the curvature is the failure of covariant derivatives to commute*, i.e. the failure of transport around an infinitesimal loop to be trivial. Under $A_\mu \rightarrow A_\mu + \partial_\mu \alpha$, $F_{\mu\nu}$ is invariant ($\partial_\mu \partial_\nu \alpha$ is symmetric). The Maxwell action is the simplest gauge-invariant, Lorentz-invariant functional:

$$S_{\text{Maxwell}} = -\frac{1}{4e^2} \int d^d x F_{\mu\nu} F^{\mu\nu}.$$

Example 2.1 (Differential-form repackaging)

Write $A = A_\mu dx^\mu$ (a 1-form) and $F = dA$ (a 2-form), so $F = \frac{1}{2} F_{\mu\nu} dx^\mu \wedge dx^\nu$. Then gauge invariance is $F = dA \rightarrow d(A + d\alpha) = dA$ because $d^2 = 0$. The Bianchi identity $dF = 0$ is automatic, and it *is* the homogeneous Maxwell equations $\nabla \cdot \mathbf{B} = 0$, $\nabla \times \mathbf{E} + \partial_t \mathbf{B} = 0$. This is your first taste of how much cleaner gauge theory looks in form language; we lean on it heavily in §4.

Exercise 2.2 ((classic) forms reproduce Maxwell) — With $A = A_\mu dx^\mu$ in 4d, verify $F = dA$ has components $F_{0i} = E_i$, $F_{ij} = -\epsilon_{ijk} B_k$. Show $dF = 0$ gives the two homogeneous Maxwell equations and $d \star F = \star j$ gives the two inhomogeneous ones. (Here $\star$ is the Hodge star.)

2.4 The Wilson line: transport made finite

The covariant derivative is the infinitesimal version of transport. Its finite version is the **Wilson line**. Along a path γ from x to y ,

$$W_\gamma = \exp\left(i \int_\gamma A\right) = \exp\left(i \int_\gamma A_\mu dx^\mu\right).$$

It is the phase you accumulate transporting a unit-charge particle along γ . Under a gauge transformation it transforms at the endpoints, $W_\gamma \rightarrow e^{i\alpha(y)} W_\gamma e^{-i\alpha(x)}$, so $\phi^*(y) W_\gamma \phi(x)$ is gauge invariant, and a **Wilson loop** $W_{\text{loop}} = \exp(i \oint A)$ around a closed γ is fully gauge invariant.

Example 2.2 (Aharonov–Bohm: the holonomy is physical)

Take a solenoid carrying flux Φ so that $\mathbf{B} = 0$ outside but $\oint A = \Phi \neq 0$ for loops encircling it. An electron beam split around the solenoid recombines with a relative phase $e^{i \oint A} = e^{i\Phi}$, shifting the interference pattern—a measurable effect even though $\mathbf{B} = 0$ along both paths.

Lesson: the gauge-invariant, physical content of a gauge field is its holonomy around loops, not its pointwise value. Hold onto this; for finite groups it becomes the *entire* definition.

Exercise 2.3 ((medium) flux quantization from single-valuedness) — A charge- q wavefunction must be single valued. Argue that on a closed two-dimensional surface the total magnetic flux must satisfy $q \oint A = q \int F \in 2\pi\mathbb{Z}$. (This is the Dirac quantization condition; we re-derive it geometrically as “ $c_1 \in \mathbb{Z}$ ” in §4.6.)

3 Non-abelian gauge fields: Yang–Mills

Replace the phase group $U(1)$ by a general Lie group G (think $SU(2)$, $SU(3)$). The structure is identical; only the bookkeeping is richer because G is non-commutative.

3.1 Fields in representations; the connection is Lie-algebra valued

Now matter is no longer a single complex number. It is a multiplet $\psi(x)$ that transforms in a **representation** R of G . Recall what that means: a representation assigns to each group element $g \in G$ a matrix $R(g)$ acting on the column vector ψ , compatibly with the group law, $R(g_1)R(g_2) = R(g_1g_2)$. A local gauge transformation is

$$\psi(x) \mapsto R(g(x)) \psi(x), \quad g(x) \in G,$$

and we abbreviate $R(g) \equiv g$ when the representation is understood. Exactly as for the scalar in §2, the ordinary derivative is not covariant: $\partial_\mu(g\psi) = g \partial_\mu\psi + (\partial_\mu g)\psi$, and the second term is the obstruction we must absorb.

First attempt, and why it leaves us unsatisfied. Mimic §2 and introduce a matrix-valued gauge field $\mathcal{A}_\mu$ of the same size as ψ , with covariant derivative $D_\mu\psi = (\partial_\mu - i\mathcal{A}_\mu)\psi$. Demanding the covariant transformation $D_\mu\psi \rightarrow g D_\mu\psi$ fixes the law of $\mathcal{A}_\mu$: expanding $(\partial_\mu - i\mathcal{A}'_\mu)g\psi \stackrel{!}{=} g(\partial_\mu - i\mathcal{A}_\mu)\psi$ gives

$$\mathcal{A}_\mu \mapsto g \mathcal{A}_\mu g^{-1} - i(\partial_\mu g)g^{-1} = g \mathcal{A}_\mu g^{-1} + i g \partial_\mu g^{-1}$$

(the two forms agree because $\partial_\mu(gg^{-1}) = 0$). This works—but it should make you uneasy, for a reason worth dwelling on. We defined $\mathcal{A}_\mu$ as a matrix in the *matter* representation. Does every multiplet—the quark triplet, an adjoint gaugino, a charged singlet—then carry its *own* gauge field? What if there is no matter at all? And how many of the entries of the matrix $\mathcal{A}_\mu$ are genuinely independent? In Maxwell theory a single photon A_μ coupled to all charged fields at once; we want that same economy here.

The resolution: one Lie-algebra-valued connection feeds every representation. The cure is to insist that, in *every* representation, $\mathcal{A}_\mu$ be assembled from the *same* set of real coefficient functions $A_\mu^a(x)$:

$$\mathcal{A}_\mu(x) = A_\mu^a(x) \tau^a, \quad \tau^a \equiv R_*(T^a),$$

where the τ^a are the generators of G in the representation R —the matrices appearing in $R(e^{i\theta^a T^a}) = e^{i\theta^a \tau^a}$ —and the T^a are the abstract generators. The honest gauge field is then the representation-*independent* object

$$A_\mu(x) = A_\mu^a(x) T^a \in \mathfrak{g},$$

valued in the **Lie algebra** $\mathfrak{g}$ of G : the tangent space to G at the identity, spanned by the generators T^a , closed under the bracket $[T^a, T^b] = i f^{abc} T^c$ (the real numbers f^{abc} are the **structure constants**, and $\exp$ maps $\mathfrak{g}$ back into G). Each matter field reconstructs the matrix it needs, $\mathcal{A}_\mu = A_\mu^a \tau^a$, from this one A_μ . We say A_μ is **Lie-algebra valued**, and it—not any of the $\mathcal{A}_\mu$ —is the gauge field.

Why this is consistent. For “ $A_\mu \in \mathfrak{g}$ ” to be a meaningful constraint, the gauge transformation must keep A_μ inside $\mathfrak{g}$, and it does—both terms cooperate. The homogeneous term $g A_\mu g^{-1}$ stays in $\mathfrak{g}$ because conjugating a generator by a group element lands back in the algebra; this is precisely the **adjoint representation** Ad_g of G acting on $\mathfrak{g}$. The inhomogeneous term lies in $\mathfrak{g}$ too: writing $g(x) = e^{i\theta^a(x) T^a}$ near a point, $i g \partial_\mu g^{-1} = \partial_\mu \theta^a T^a + \dots$, a Lie-algebra element. So the gauge group acts *within* $\mathfrak{g}$. *This is the real reason the connection is Lie-algebra valued*: it is the unique representation-independent carrier of the gauge field, and gauge transformations are exactly the operations that preserve that property. Collecting the result,

$$D_\mu \psi = (\partial_\mu - i \mathcal{A}_\mu) \psi, \quad \boxed{A_\mu \mapsto g A_\mu g^{-1} + i g \partial_\mu g^{-1}}$$

with the law chosen so that $D_\mu \psi \rightarrow g D_\mu \psi$, just as in §2. Setting $G = \text{U}(1)$, $g = e^{i\alpha}$ collapses this to $A_\mu \rightarrow A_\mu + \partial_\mu \alpha$: the abelian case is simply the special case with one generator, where conjugation does nothing.

Note 3.1 (Two objects, easy to conflate: A_μ versus $\mathcal{A}_\mu$). The connection $A_\mu = A_\mu^a T^a$ lives in the abstract Lie algebra $\mathfrak{g}$ and knows nothing about any matter field. The matrix $\mathcal{A}_\mu = A_\mu^a \tau^a$ that sits inside a covariant derivative lives in whatever representation R the matter occupies. Think of A_μ as *the* gauge field and $\mathcal{A}_\mu$ as “how it looks” when handed to a particular multiplet. Physicists write both as “ A_μ ” and let context disambiguate; we will too, but keep the distinction in your pocket—it is the difference between an element of $\mathfrak{g}$ and its image under a representation.^a

^aThe argument that one $\mathfrak{g}$ -valued field serves all representations, and the check that gauge transformations preserve $\mathfrak{g}$, follows Harlow, *8.325 Quantum Field Theory III* lecture notes, §3.1.

Exercise 3.1 ((easy) the inhomogeneous term really lands in $\mathfrak{g}$) — We claimed $i g \partial_\mu g^{-1} \in \mathfrak{g}$. Prove it: differentiate $g g^{-1} = \mathbf{1}$ to get $g \partial_\mu g^{-1} = -(\partial_\mu g) g^{-1}$, then write $g(x) = e^{i\theta^a(x) T^a}$ near a point and expand to first order to show $i g \partial_\mu g^{-1} = \partial_\mu \theta^a T^a + \dots \in \mathfrak{g}$. *This is the “due process” behind the claim that gauge transformations keep A_μ Lie-algebra valued.*

Exercise 3.2 ((medium) the connection is an affine, not linear, object) — Show that if A_μ and A'_μ are two connections then their difference $A_\mu - A'_\mu$ transforms in the adjoint representation (homogeneously, no ∂g term). Conclude that connections form an affine space modeled on $\mathfrak{g}$ -valued 1-forms. *This is why “the difference of two gauge fields is a matter field” and why the space of connections is flat (affine) but not a vector space.*

3.2 Curvature and the Yang–Mills action

The field strength now has an extra, crucial nonlinear piece. The cleanest definition is still the commutator of covariant derivatives, $F_{\mu\nu} = i[D_\mu, D_\nu]$, which works out to

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu - i[A_\mu, A_\nu], \quad F = dA - iA \wedge A,$$

or, in components, $F_{\mu\nu}^a = \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + f^{abc} A_\mu^b A_\nu^c$. A word on the notation $A \wedge A$: it means $A_\mu A_\nu dx^\mu \wedge dx^\nu$, the matrix product of the coefficients wedged together. Unlike the wedge of an *ordinary* 1-form with itself, this does *not* vanish—the matrices A_μ and A_ν need not commute, and indeed $A \wedge A = \frac{1}{2}[A, A]$ encodes exactly that non-commutativity. (We make this notation precise in §4.2.) Under a gauge transformation the inhomogeneous pieces cancel and

$$F_{\mu\nu} \mapsto g F_{\mu\nu} g^{-1},$$

so F is no longer gauge *invariant* but gauge *covariant* (it transforms in the adjoint). Note the sharp contrast with Maxwell: F is gauge-invariant if and only if the adjoint representation is trivial—i.e. if and only if G is abelian. For non-abelian G we must take a trace to build an invariant action,

$$S_{\text{YM}} = -\frac{1}{2g^2} \int d^d x \operatorname{tr}(F_{\mu\nu} F^{\mu\nu}) = -\frac{1}{4g^2} \int d^d x F_{\mu\nu}^a F^{a\mu\nu},$$

the second form using a basis with $\operatorname{tr}(T^a T^b) = \frac{1}{2}\delta^{ab}$. The coupling g can be slid into the covariant derivative by rescaling $A \rightarrow gA$ (then $D_\mu = \partial_\mu - igA_\mu$ and the kinetic term is canonically $-\frac{1}{4}F^a F^a$); nothing physical depends on where it lives.

Intuition 3.1 (Why gauge bosons self-interact)

The $-i[A_\mu, A_\nu]$ term in F means $\operatorname{tr} F^2$ contains cubic and quartic terms in A : gluons carry color charge and interact with each other, unlike photons. The deep reason is non-commutativity of G . “Non-abelian” is the whole difference between electromagnetism (a free Maxwell field is a free theory) and QCD (a free Yang–Mills field is already an interacting, confining theory).

Example 3.1 (The Standard Model connection)

The SM gauge group is $G = \text{SU}(3)_c \times \text{SU}(2)_L \times \text{U}(1)_Y$. The gluon A_μ is an $\text{SU}(3)$ connection (8 components A_μ^a , $a = 1, \dots, 8$); W_μ, Z_μ, γ come from the $\text{SU}(2) \times \text{U}(1)$ connection after the Higgs mechanism reshuffles them. Quarks ψ live in the fundamental **3** of color; “minimal coupling” $D_\mu = \partial_\mu - ig_s A_\mu$ is precisely how quarks talk to gluons. **Everything you know about forces is the statement that these connections are dynamical.**

3.3 Wilson lines, holonomy, and path ordering

Because A_μ at different points need not commute, the finite transport requires **path ordering**:

$$W_\gamma = \mathcal{P}\exp\left(i \int_\gamma A_\mu dx^\mu\right), \quad W_{\text{loop}} = \text{tr } \mathcal{P}\exp\left(i \oint_\gamma A\right).$$

W_γ is the holonomy: the group element by which the internal frame rotates after being transported around γ . The trace of a Wilson loop is gauge invariant and is *the* basic observable in pure gauge theory (it is the order parameter for confinement; see §9).

Exercise 3.3 ((thinker) curvature is infinitesimal holonomy) — Show that the holonomy of A around an infinitesimal square loop of side ϵ in the $\mu\nu$ -plane is $W = \mathbf{1} - i\epsilon^2 F_{\mu\nu} + \mathcal{O}(\epsilon^3)$. Conclude that $F_{\mu\nu}$ measures holonomy per unit area, and that “flat connection” ($F = 0$) means “transport depends only on the homotopy class of the path.” *Hint: multiply four Wilson lines along the edges and use Baker–Campbell–Hausdorff.*

Note 3.2 (Two conventions you will meet). Physicists use Hermitian generators and $D = \partial - iA$, giving $F = dA - iA \wedge A$ and $A \rightarrow gAg^{-1} + ig dg^{-1}$. Mathematicians use anti-Hermitian generators and $D = d + A$, giving $F = dA + A \wedge A$ and $A \rightarrow gAg^{-1} - (dg)g^{-1}$. They differ only by where the i ’s sit ($A_{\text{math}} = -iA_{\text{phys}}$). We use the physics convention throughout these notes; just know that bundle-theory texts drop the i ’s (and flip the matching signs) without changing any physics.

4 The geometric definition: connections on principal bundles

So far $A_\mu(x)$ has been a single field on spacetime. This is a lie—a useful, local lie. Globally, a gauge field is generally *not* a single $\mathfrak{g}$ -valued 1-form, and pretending otherwise misses monopoles, instantons, and the entire topological side of gauge theory. The honest definition is geometric. Following the Seiberg/Witten attitude you asked for, we develop exactly as much bundle theory as buys us real understanding, and not more.

4.1 Why the local picture is not enough: the Dirac monopole

Example 4.1 (The monopole forces you into bundles)

A magnetic monopole at the origin has $\mathbf{B} = \frac{g_m}{2} \frac{\hat{r}}{r^2}$, so $\int_{S^2} F = 4\pi \cdot \frac{g_m}{2} \neq 0$ on a sphere surrounding it. But if $F = dA$ *globally* with a single smooth A , then Stokes gives $\int_{S^2} dA = 0$. Contradiction. The resolution (Wu–Yang): cover S^2 with two patches (north/south), use a different potential A_N, A_S on each,

$$A_N - A_S = d\lambda \quad \text{on the overlap (the equator),}$$

i.e. the two are related by a *gauge transformation* where they overlap. The “field” is really the pair (A_N, A_S) glued by a transition function $g_{NS} = e^{i\lambda}$. Single-valuedness of g_{NS} forces $\frac{1}{2\pi} \int_{S^2} F \in \mathbb{Z}$: the monopole charge is quantized and *topological*. There is no global A ; there is

a **bundle**.^a

^aThis is the Wu–Yang construction, *Phys. Rev. D* **12** (1975); a modern pedagogical version (patchwork gauge fields, Dirac quantization) is Tong, *Lectures on Gauge Theory* §1.1.

Let us draw the lesson slowly, because it is the whole point of the section. On the sphere S^2 surrounding the monopole there is simply *no* smooth global vector potential: any single A has a **Dirac string**, a line of singularity where the gauge choice breaks down (heuristically, the return flux of an infinitely thin solenoid feeding the monopole). The Wu–Yang cure is to use two potentials— A_N , regular everywhere except the south pole, and A_S , regular everywhere except the north pole—each perfectly good on its own patch (the northern and southern hemispheres, which overlap in a band around the equator). Where they overlap they describe the *same* magnetic field, so they can differ only by a gauge transformation, $A_N = A_S + d\lambda$. The phase $g_{NS} = e^{i\lambda} : S_{\text{eq}}^1 \rightarrow \text{U}(1)$ is the **transition function** gluing the two descriptions. For a charged wavefunction to be single valued, g_{NS} must be single valued around the equator, so λ winds by an integer multiple of 2π ; that integer is the monopole charge, and it equals $\frac{1}{2\pi} \int_{S^2} F$. *The non-existence of a global A , and the integer measuring it, are topological facts about how the patches are glued—not about any local field value.*

This is what a gauge field is, told honestly: a collection of local potentials $\{A_i\}$ on patches $\{U_i\}$ covering spacetime, together with transition functions $\{g_{ij}\}$ saying how to pass between patches. The geometric object that packages this data is a **connection on a principal bundle**. We build it in stages—notation, then the bundle, then matter, then the connection and its curvature—giving each piece its due.

4.2 Notation interlude: Lie-algebra-valued forms

We have been writing $A = A_\mu^a T^a dx^\mu$ and $F = dA - iA \wedge A$ without saying exactly what kind of object these are. Since the rest of the section leans on it, let us fix it now. Write $\Omega^k(M)$ for the space of ordinary differential k -forms on M .

Definition 4.1 (Lie-algebra-valued forms)

A **$\mathfrak{g}$ -valued k -form** is an object $\alpha = \alpha^a T^a$ whose coefficients $\alpha^a \in \Omega^k(M)$ are ordinary k -forms; their space is written $\Omega^k(M, \mathfrak{g})$. Thus the gauge field $A \in \Omega^1(M, \mathfrak{g})$ and the field strength $F \in \Omega^2(M, \mathfrak{g})$. For two such forms one wedges the form parts while bracketing the algebra parts,

$$[\alpha, \beta] \stackrel{\text{def}}{=} \alpha^a \wedge \beta^b [T^a, T^b], \quad \text{so in particular} \quad A \wedge A \stackrel{\text{def}}{=} A_\mu A_\nu dx^\mu \wedge dx^\nu = \frac{1}{2} [A, A].$$

Two points repay attention, both already lurking in §3. First, $A \wedge A \neq 0$ even though the wedge of an ordinary 1-form with itself vanishes: here the coefficients are non-commuting matrices, and $A \wedge A = \frac{1}{2} [A, A]$ is precisely their failure to commute. Second, the natural derivative on $\mathfrak{g}$ -valued forms, once a connection is present, is the **covariant exterior derivative**

$$d_A \alpha \stackrel{\text{def}}{=} d\alpha - i[A, \alpha],$$

which is to d what D_μ is to ∂_μ . With it, $F = dA - iA \wedge A$ and the Bianchi identity is just $d_A F = 0$. Throughout, $\text{tr}(\cdots)$ denotes the trace over the matrix (representation) indices; that trace is what turns the covariant $\text{tr}(F \wedge F)$ into an ordinary, gauge-invariant form.

4.3 Principal G -bundles

Definition 4.2 (Principal G -bundle)

A **principal G -bundle** over spacetime M is a manifold P with a free, fiber-preserving right action of G and a projection $\pi : P \rightarrow M$ such that locally P looks like a product: there is a cover $\{U_i\}$ of M and diffeomorphisms (**local trivializations**) $\Phi_i : \pi^{-1}(U_i) \xrightarrow{\sim} U_i \times G$ commuting with the G -action. On overlaps $U_i \cap U_j$ the two trivializations differ by **transition functions**

$$g_{ij} : U_i \cap U_j \rightarrow G, \quad g_{ij}g_{jk} = g_{ik} \quad (\text{cocycle condition}).$$

Every clause earns its keep; let us unpack them.

- The **total space** P sits above spacetime via the **projection** $\pi : P \rightarrow M$. The **fiber** over $x \in M$ is $\pi^{-1}(x)$, the set of points of P lying above x . (The notation $\pi^{-1}(U_i)$ in the definition means the part of P sitting above the patch U_i .)
- “ G acts freely and fiber-preservingly on the right” means G moves points within each fiber with no fixed points. A free transitive action makes each fiber a **torsor** for G : a copy of G that has forgotten which element is the identity. Physically the fiber is the space of internal frames at x , with no canonical “unrotated” frame—choosing one is choosing a gauge.
- A **local trivialization** $\Phi_i : \pi^{-1}(U_i) \xrightarrow{\sim} U_i \times G$ is a choice, over the patch U_i , of how to identify each fiber with G itself; the symbol “ $\xrightarrow{\sim}$ ” denotes such an identifying (diffeomorphic) map. It always exists locally—the whole content of the bundle is whether the local choices can be made to agree everywhere at once.
- Comparing two trivializations on an overlap gives the **transition function** $g_{ij} = \Phi_i \circ \Phi_j^{-1}$. Consistency on triple overlaps forces the **cocycle condition** $g_{ij}g_{jk} = g_{ik}$. The g_{ij} carry all the topology; relabeling each patch, $g_{ij} \rightarrow h_i g_{ij} h_j^{-1}$, describes the same bundle.

A **section** is a map $s : M \rightarrow P$ with $\pi \circ s = \text{id}_M$ —a choice of one point in each fiber, that is, a global gauge. A principal bundle admits a global section if and only if it is trivial, $P \cong M \times G$. This is the precise geometric content of “you cannot always fix a gauge globally”: a nontrivial bundle has no global section, only local ones s_i over each U_i .

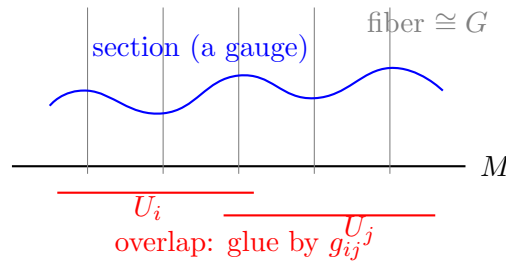
Intuition 4.1 (What the bundle “is,” physically)

The fiber over $x \in M$ is a copy of G : it is the space of all possible internal frames (gauge choices) at x . A **gauge** is a choice of section—a way of picking one frame at each point—and it exists globally only if the bundle is trivial. The transition functions g_{ij} are the gauge transformations relating the conventions of neighboring patches. *The topology of how the fibers are glued is exactly the topology that monopole charge and instanton number measure.* Matter fields live in **associated bundles**: replace the fiber G by the representation space V on which G acts, and ψ becomes a section of $P \times_G V$ (defined carefully in §4.4).

Example 4.2 (The simplest nontrivial bundle: cylinder vs. Möbius band)

Take $M = S^1$ and let $G = \mathbb{Z}_2 = \{\pm 1\}$ act on a line fiber by sign. Cover the circle with two arcs, so there are two overlap regions. Over each arc the bundle is just a product (a short ribbon); the only freedom is the transition function $\in \mathbb{Z}_2$ on each overlap. If both gluings are $+1$ you get a **cylinder**—the trivial bundle, on which a global nonvanishing section exists. If one gluing is -1 you get a **Möbius band**: going once around S^1 flips the fiber, and no global nonvanishing section exists. The invariant distinguishing the two is exactly the holonomy $\in \mathbb{Z}_2$ around the circle. *That is your first “ $\mathbb{Z}_2$ gauge field,” met as honest geometry; all of §5 is this picture with S^1 replaced by a general spacetime and $\mathbb{Z}_2$ by any group.*

A cartoon of the local picture (base, fibers, a section, and overlapping trivializations):

**Example 4.3** (A small zoo of bundles)

(a) Trivial bundle $P = M \times G$: a global gauge exists; this is the case implicitly assumed in §2–§3. (b) The Hopf bundle $U(1) \hookrightarrow S^3 \rightarrow S^2$: the charge-1 Dirac monopole. Its transition function on the S^2 equator winds once around $U(1)$. (c) A flat G -bundle: transition functions can be taken *constant*, and the bundle is classified by a homomorphism $\pi_1(M) \rightarrow G$ (we use this constantly for finite G in §5). (d) The frame bundle of M : structure group $GL(d)$ (or $SO(d)$ with a metric); its connection is the Levi-Civita connection of GR.

4.4 Associated bundles: where matter lives

A principal bundle is about the gauge group itself. Matter fields live in **associated bundles** built from it, and spelling this out finally explains the notation $P \times_G V$ used above. Given a representation (R, V) of G (so V is the vector space the multiplet lives in), form the **associated vector bundle**

$$E = P \times_G V \stackrel{\text{def}}{=} (P \times V)/\sim, \quad (p \cdot g, v) \sim (p, R(g)v).$$

In words: take pairs (frame p , vector v) and declare that rotating the frame by g while rotating the vector by $R(g)$ gives the same physical point. A **matter field** in representation R is then a **section** of E : in a local gauge it is an ordinary V -valued function ψ_i on U_i , and on overlaps the two descriptions are related by

$$\psi_i = R(g_{ij}) \psi_j \quad \text{on } U_i \cap U_j.$$

This is the global meaning of the rule “ $\psi \rightarrow g\psi$ ” from §3: a gauge transformation is the transition function g_{ij} acting in the representation R . The covariant derivative $D_\mu \psi$ is the instruction for

differentiating such a section in a way that respects the gluing—which is exactly what the connection provides, next. Finally, the curvature F will live in the associated bundle for the *adjoint* representation, the **adjoint bundle** $\text{ad } P = P \times_G \mathfrak{g}$; that is what the phrases “ F transforms in the adjoint” and “ F is valued in $\text{ad } P$ ” mean precisely.

4.5 Connections: the gauge field as a global object

What does a connection *do*? It provides **parallel transport**: a consistent rule for lifting a path in the base M to a path in the total space P , i.e. for carrying an internal frame along as you move through spacetime. Infinitesimally, this is a choice at each point of P of which directions count as **horizontal**—the directions “along which the frame is not rotating.” That choice is encoded in a single $\mathfrak{g}$ -valued 1-form on P .

Definition 4.3 (Connection / gauge field)

A **connection** on P is a $\mathfrak{g}$ -valued 1-form $\omega \in \Omega^1(P, \mathfrak{g})$ on the total space, subject to two conditions:

- (i) on vertical (along-the-fiber) directions ω returns the Lie-algebra element generating them, so the horizontal directions are exactly $\ker \omega$;
- (ii) it is G -equivariant, $R_g^* \omega = \text{Ad}_{g^{-1}} \omega$ (here R_g is the right action of g and R_g^* its pullback), so the horizontal directions are carried consistently around each fiber.

This is the mathematician’s (Ehresmann) definition, and it deliberately looks nothing like the physicist’s A_μ . *The bridge between them is pullback by a section.* Pick a local gauge, i.e. a section $s_i : U_i \rightarrow P$. Then

$$A_i \stackrel{\text{def}}{=} s_i^* \omega \in \Omega^1(U_i, \mathfrak{g})$$

is an honest $\mathfrak{g}$ -valued 1-form on the patch U_i —and *this is the physicist’s* $A_\mu^a T^a dx^\mu$. (“Pullback s_i^* ” just means: ω lives on P , the section s_i maps U_i into P , and we read ω back along that map to get a form on U_i .) The abstract ω is gauge-independent; the familiar A_μ is what you see *after* choosing a gauge s_i . Two sections over an overlap differ by a fiber rotation, $s_j = s_i \cdot g_{ij}$, and pulling back ω along each yields the **gluing law**

$$A_j = g_{ij}^{-1} A_i g_{ij} + i g_{ij}^{-1} dg_{ij}$$

This is *identically* the gauge transformation of §3, now with a sharper meaning: it is how the one connection ω looks in two different local gauges. Changing the section on a single patch, $s_i \rightarrow s_i \cdot h(x)$, is an ordinary gauge transformation; passing between patches is the transition g_{ij} ; the gluing law shows these are the *same* operation. That is why “gauge redundancy” and “patching data” are one notion, not two.

Equation 4.1 (The definition to carry in your head)

A G -gauge field on M = a connection on a principal G -bundle over M .

On each patch it is the familiar $A_\mu^a T^a dx^\mu$; globally it is the patches *plus* the gluing data g_{ij} .

Two gauge fields are physically the same iff related by a (globally defined) gauge transformation, i.e. a bundle automorphism.

The finite version of parallel transport is the **holonomy**: lift a loop γ horizontally and read off the group element relating the starting frame to the ending one. In a local gauge this is exactly the path-ordered Wilson line of §3, $\text{Hol}(\gamma) = \mathcal{P}\exp i \oint_{\gamma} A$. So the Wilson loop is not an extra gadget bolted onto gauge theory: it *is* the connection's parallel transport, written in a gauge.

Definition 4.4 (Curvature)

The **curvature** of the connection is the $\mathfrak{g}$ -valued 2-form

$$F = dA - i A \wedge A = dA - \frac{i}{2}[A, A]$$

(physics convention, matching §3; the two forms agree since $A \wedge A = \frac{1}{2}[A, A]$). Although the local potentials A_i glue by the inhomogeneous law above, the curvature glues *homogeneously*, $F_j = g_{ij}^{-1} F_i g_{ij}$, with no dg term. Hence F is a genuinely globally defined 2-form valued in the adjoint bundle $\text{ad } P$. It obeys the **Bianchi identity** $d_A F \equiv dF - i[A, F] = 0$ (for abelian G this is just $dF = 0$, the homogeneous Maxwell equations). Geometrically, F measures the failure of horizontal lifts to close up: the holonomy around an infinitesimal loop in the $\mu\nu$ -plane is $\mathbf{1} - iF_{\mu\nu} dx^\mu dx^\nu + \dots$. This is the precise sense, promised in §3.3, in which *curvature is infinitesimal holonomy*.

Exercise 4.1 ((medium) the inhomogeneous term cancels in F) — Starting from the gluing law for A , verify directly that $F = dA - iA \wedge A$ glues covariantly, $F_j = g_{ij}^{-1} F_i g_{ij}$, with no dg term. *This is the calculation that explains why $\text{tr } F^k$ are globally defined and give topological invariants.*

4.6 Topology: characteristic classes, monopoles, instantons

Because $\text{tr } F^k$ is globally defined, gauge invariant, and closed (closed by the Bianchi identity), its integral over a closed cycle does not change under any continuous deformation of the connection: it is a **topological invariant** of the bundle. The systematic recipe for manufacturing such invariants from F is **Chern–Weil theory**, and the resulting de Rham classes are the **characteristic classes**. Two of them run most of physics.

Equation 4.2 (The two invariants you must know)

First Chern number (abelian flux / monopole charge), over a closed surface Σ :

$$c_1 = \frac{1}{2\pi} \int_{\Sigma} F \in \mathbb{Z}.$$

Instanton number / second Chern number (non-abelian, over a closed 4-manifold X):

$$k = \frac{1}{8\pi^2} \int_X \text{tr}(F \wedge F) \in \mathbb{Z}.$$

Integrality is the geometric origin of Dirac charge quantization and of the quantization of the θ -angle's periodicity.

Why are these *integers*, rather than arbitrary real numbers? Both count windings. For the abelian c_1 , cover Σ by patches; on overlaps the potentials differ by $d\lambda_{ij}$, and applying Stokes patch by patch reduces $\frac{1}{2\pi} \int_{\Sigma} F$ to the total winding of the transition functions $e^{i\lambda_{ij}} : S^1 \rightarrow U(1)$ around the patch boundaries. A map $S^1 \rightarrow U(1)$ has an integer winding number—this is the statement $\pi_1(U(1)) = \mathbb{Z}$ —so $c_1 \in \mathbb{Z}$. This is precisely the monopole quantization of §4.1, now recognized as a fact about π_1 of the gauge group. For the non-abelian instanton number, a finite-action field on $\mathbb{R}^4$ must approach pure gauge at infinity, $A \rightarrow ig dg^{-1}$, defining a map $g : S_{\infty}^3 \rightarrow G$; its winding lives in $\pi_3(G) = \mathbb{Z}$ for every simple G , and equals $\frac{1}{8\pi^2} \int \text{tr}(F \wedge F)$. Here $\pi_n(X)$ denotes the n -th **homotopy group**, the set of homotopy classes of maps $S^n \rightarrow X$ —the precise bookkeeping of “how many times one sphere wraps a space.”

Example 4.4 (Instantons)

On $\mathbb{R}^4$ with finite action, A must approach pure gauge at infinity, $A \rightarrow ig dg^{-1}$ with $g : S_{\infty}^3 \rightarrow G$. For $G = \text{SU}(2) \cong S^3$ such maps are classified by $\pi_3(\text{SU}(2)) = \mathbb{Z}$, the instanton number k above. An instanton is a finite-action solution in a fixed topological sector $k \neq 0$; it tunnels between topologically distinct vacua and underlies the θ -vacuum structure and the resolution of the $U(1)_A$ problem.

Exercise 4.2 ((thinker) θ -term is topological) — Show that $\frac{\theta}{8\pi^2} \int_X \text{tr}(F \wedge F)$ adds θk to the action with $k \in \mathbb{Z}$, so the path integral e^{iS} is periodic, $\theta \sim \theta + 2\pi$. Explain why this term does not affect the classical equations of motion (it is a total derivative / depends only on the bundle), yet it has real physical effects nonperturbatively. *This is your first “coupling constant that is really a topological label.”*

Note 4.1 (How a practitioner actually uses bundles). You rarely manipulate ω on the total space P . In practice: (1) work locally with A_{μ}^a as in §3; (2) remember that globally there may be several patches glued by g_{ij} , which is where topology lives; (3) extract topology from gauge-invariant integrals $\frac{1}{2\pi} \int F$, $\frac{1}{8\pi^2} \int \text{tr} F^2$, and from holonomies $\oint A$. The bundle language is the *bookkeeping that guarantees these are well-defined*; you reach for it precisely when something topological is at stake (monopoles, instantons, flux sectors, anomalies, finite gauge groups).

Preview 4.1 (Where this is heading)

The phrase “ A is a $U(1)$ gauge field” now has a precise meaning: A is a connection on a principal $U(1)$ -bundle, equivalently the collection of local potentials A_i with $U(1)$ -valued transition functions, with gauge-invariant content encoded in holonomies $\oint A$ and Chern numbers $\frac{1}{2\pi} \int F$. In §5 we replace the continuous group by a finite one and discover that “connection” still makes perfect sense—it just becomes an assignment of group elements to paths.

5 Finite-group gauge fields: what is a $\mathbb{Z}_2$ -gauge field?

Here is the puzzle that sends most students reeling the first time. A finite group like $\mathbb{Z}_2$ has *no Lie algebra*: there is no $\mathfrak{g}$, no $A_\mu^a T^a$, no continuous $A_\mu(x)$. So what on earth does “ μ is a $\mathbb{Z}_2$ -gauge field” mean? The answer is that we have been overcomplicating “connection.” Strip it to its essence.

5.1 A connection is a rule for parallel transport along paths

Intuition 5.1 (The essence of a connection)

Forget A_μ . A connection is, at heart, a recipe that assigns to each path $\gamma : x \rightarrow y$ a group element $\text{Hol}(\gamma) \in G$ —the holonomy—such that **(i)** composing paths multiplies holonomies, $\text{Hol}(\gamma_2\gamma_1) = \text{Hol}(\gamma_2)\text{Hol}(\gamma_1)$, and **(ii)** reversing a path inverts it. For a Lie group this recipe is generated by integrating A_μ ($\text{Hol} = \mathcal{P}\exp i\int A$). But the recipe makes sense for *any* group G , including a finite one—we just specify the group elements directly. **That is a G -gauge field.**

For a finite group there is no continuous local data to integrate; the connection lives on the discrete skeleton of spacetime. So we put spacetime on a lattice (or a triangulation / simplicial complex) and assign the data to edges.

5.2 The lattice definition: link variables

Definition 5.1 (Lattice G -gauge field, G any group)

Triangulate (or put a lattice on) spacetime, with vertices, edges, and plaquettes (2-cells). A **G -gauge field** is an assignment of a group element to each oriented edge e ,

$$g_e \in G, \quad g_{\bar{e}} = g_e^{-1},$$

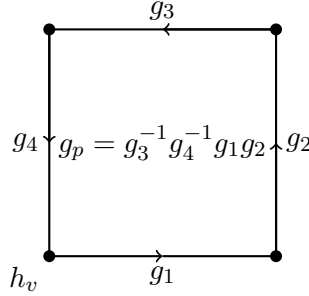
i.e. the holonomy across that link. A **gauge transformation** is a choice of group element $h_v \in G$ at each vertex, acting by

$$g_e \mapsto h_{t(e)} g_e h_{s(e)}^{-1} \quad (s, t = \text{source, target of } e).$$

The **field strength** (curvature) is the holonomy around each plaquette p ,

$$g_p = \prod_{e \in \partial p} g_e \quad (\text{ordered product around the boundary}),$$

which transforms by conjugation $g_p \rightarrow h g_p h^{-1}$. The configuration is **flat** if $g_p = \mathbf{1}$ for every plaquette.

**Intuition 5.2** (This is literally the continuum story, discretized)

Edge variable $g_e \leftrightarrow$ Wilson line $\mathcal{P}\exp i\int_e A$ across a link. Vertex transformation $h_v \leftrightarrow$ gauge function $g(x)$. Plaquette product $g_p \leftrightarrow \exp(i\oint_{\partial p} A) = \exp(iF \cdot \text{area})$, the infinitesimal-loop holonomy from the Exercise in §3.3. “Flat” $\leftrightarrow$ “ $F = 0$.” Nothing new conceptually—we just dropped the requirement that G be continuous, and the integrals became products.

Example 5.1 ($\mathbb{Z}_2$ gauge field, concretely)

Take $G = \mathbb{Z}_2 = \{+1, -1\}$ (multiplicatively) or $\{0, 1\}$ with addition mod 2. A $\mathbb{Z}_2$ gauge field assigns $g_e = \pm 1$ to each edge. A gauge transformation flips $g_e \rightarrow h_t g_e h_s$ with $h_v = \pm 1$ at vertices. The plaquette curvature is the product of the four edge signs around a square; flat means every plaquette product is $+1$. This is exactly Wegner’s $\mathbb{Z}_2$ lattice gauge theory (a.k.a. Ising gauge theory), and the dynamical version is the deconfined phase that is the toric code.^a When someone writes “ μ is a $\mathbb{Z}_2$ -gauge field,” μ is precisely this edge assignment (often written additively, $\mu \in C^1(M; \mathbb{Z}_2)$; see below).

^aLattice gauge theory of finite groups: Kogut, *Rev. Mod. Phys.* **51** (1979). For the $\mathbb{Z}_2$ /toric-code/condensed-matter side of everything in this section, McGreevy, *Generalized Symmetries in Condensed Matter*, [arXiv:2204.03045](#); and Harlow, *8.325 (QFT III)* §8.1 for a clean Hamiltonian lattice answer to “what is a gauge field?”

Exercise 5.1 ((easy) counting $\mathbb{Z}_2$ configurations) — On a single square (4 edges, 4 vertices, 1 plaquette), count: (a) the number of $\mathbb{Z}_2$ gauge field configurations; (b) the number of gauge transformations; (c) the number of *flat* configurations up to gauge. *Hint for (c): you should find that on the disk every flat connection is gauge-trivial.*

5.3 The cohomological repackaging (why people say “cocycle”)

For an abelian finite group like $\mathbb{Z}_N$, write the group additively. Then a gauge field is a function a from edges to $\mathbb{Z}_N$: a **1-cochain**, $a \in C^1(M; \mathbb{Z}_N)$. The plaquette curvature is $(da)_p = \sum_{e \in \partial p} a_e$, the **coboundary**. A gauge transformation shifts a by $d\lambda$ for a 0-cochain λ (the vertex data). So the picture is a perfect discrete analogue of $A \rightarrow A + d\lambda$, $F = dA$:

Equation 5.1 (Continuum vs. finite, side by side)

	continuous $U(1)$	finite $\mathbb{Z}_N$
gauge field	$A \in \Omega^1(M)$	$a \in C^1(M; \mathbb{Z}_N)$
gauge transf.	$A \rightarrow A + d\lambda$	$a \rightarrow a + d\lambda$
field strength	$F = dA$	da
flat	$dA = 0$	$da = 0$
flat mod gauge	$H^1(M; \mathbb{R})/\mathbb{Z} \cong \text{Hom}(\pi_1, U(1))$	$H^1(M; \mathbb{Z}_N) \cong \text{Hom}(\pi_1, \mathbb{Z}_N)$

Definition 5.2 (Flat finite gauge field = cohomology class = representation of π_1)

A **flat** G -gauge field ($g_p = \mathbf{1}$ for all p) up to gauge equivalence is the same data as a homomorphism $\rho : \pi_1(M) \rightarrow G$ up to conjugation. For abelian G this is $H^1(M; G) = \text{Hom}(\pi_1(M), G)$. In words: *a flat finite gauge field is completely determined by the group elements you pick up going around the non-contractible loops of spacetime.*

Intuition 5.3 (Finite gauge fields are pure holonomy)

For a continuous group, A has local wiggles (curvature) *and* global holonomies. For a finite group there is no local wiggle room—the natural configurations are flat, and *all* the information is the holonomy around non-contractible cycles. This is why finite gauge fields are intrinsically topological objects, and why $\mathbb{Z}_2$ gauge fields show up everywhere topology matters (spin structures, SPT phases, discrete θ -angles, symmetry defects).

Example 5.2 (A flat $\mathbb{Z}_2$ field on a circle / torus)

On S^1 : $\pi_1 = \mathbb{Z}$, so $\text{Hom}(\mathbb{Z}, \mathbb{Z}_2) = \mathbb{Z}_2$. There are exactly two flat $\mathbb{Z}_2$ gauge fields up to gauge: “trivial” (holonomy $+1$ around the circle) and “nontrivial” (holonomy -1). On the torus T^2 : $\pi_1 = \mathbb{Z}^2$, so there are $2 \times 2 = 4$ classes, labelled by the holonomies around the two cycles. *These four sectors are exactly the four spin structures on the torus, and the four “twisted boundary condition” sectors you sum over when gauging a $\mathbb{Z}_2$ symmetry (§7).*

Example 5.3 (Spin structures and $(-1)^F$: a $\mathbb{Z}_2$ gauge field you have used)

A choice of periodic vs. antiperiodic boundary conditions for a fermion around a cycle *is* a flat $\mathbb{Z}_2$ background for fermion parity $(-1)^F$. The “ $-$ ” in the NS sector, the Matsubara antiperiodicity of thermal fermions, the spin structure on a Riemann surface—these are all the statement “turn on a nontrivial $\mathbb{Z}_2$ gauge field.” You have been using $\mathbb{Z}_2$ gauge fields since your first thermal field theory computation without calling them that.

Exercise 5.2 ((classic) flat $\mathbb{Z}_N$ fields on surfaces) — Compute the number of gauge-inequivalent flat $\mathbb{Z}_N$ gauge fields on (a) the sphere S^2 , (b) the torus T^2 , (c) a genus- h surface Σ_h . *Answers in terms of N and h ; relate to $H^1(\Sigma_h; \mathbb{Z}_N)$.*

Exercise 5.3 ((thinker) discrete Wilson lines and non-abelian holonomy) — For $G = S_3$ (non-abelian), explain why “flat connections up to gauge” is $\text{Hom}(\pi_1, S_3)/\text{conjugation}$ rather than H^1 . On T^2 , this means pairs of commuting elements (a, b) up to simultaneous conjugation. Enumerate them. *This is the configuration sum for finite non-abelian gauge theory (Dijkgraaf–Witten).*^a

^aDijkgraaf & Witten, *Topological gauge theories and group cohomology*, *Comm. Math. Phys.* **129** (1990)—finite gauge theory and the origin of the discrete theta term as a class in $H^d(BG; \mathbb{U}(1))$.

Note 5.1 (“ μ is a $\mathbb{Z}_2$ gauge field” — decoded). When a paper introduces “a $\mathbb{Z}_2$ gauge field μ ,” it means: a $\mathbb{Z}_2$ -valued 1-cochain (assignment of ± 1 to links / a class in H^1), defined up to $\mu \rightarrow \mu + d\lambda$. If μ is *background*, it is a fixed choice probing a $\mathbb{Z}_2$ symmetry (twisted boundary conditions / a network of $\mathbb{Z}_2$ symmetry defects, §6). If μ is *dynamical*, you sum over all its classes (§7); that is $\mathbb{Z}_2$ gauge theory. The same symbol, two completely different roles—which is exactly the background-vs-dynamical distinction we turn to now.

6 Background gauge fields: probing a global symmetry

We now reach the conceptual heart of the modern point of view. We have two kinds of gauge field, distinguished not by their geometry (both are connections) but by their *role in the path integral*.

6.1 Coupling a global symmetry to a background

Suppose a theory has a global symmetry G with Noether current j^μ (continuous case) or symmetry operators U_g (general case). The defining move of the modern viewpoint:

Definition 6.1 (Background gauge field)

To **couple the symmetry G to a background gauge field** is to introduce a *fixed, non-dynamical* G -connection A on spacetime and deform the theory so that the global symmetry is promoted to a local one *with A transforming as a connection*, but **without integrating over A** . The partition function becomes a functional of the background,

$$Z[A] = \int \mathcal{D}(\text{fields}) e^{iS[\text{fields}; A]},$$

and we keep A as an external knob. Concretely, for a continuous symmetry one adds $\int A_\mu j^\mu + \mathcal{O}(A^2)$ (minimal coupling); for a finite symmetry one inserts a network of symmetry defects (§6.3).

Intuition 6.1 (A background field is a knob, a source, a probe)

Setting $A = 0$ recovers the original theory. Turning on $A \neq 0$ lets you ask: how does the theory respond to a position-dependent symmetry transformation? It is the same idea as turning on a background electromagnetic field to probe a material, or an external metric $g_{\mu\nu}$ to probe the stress tensor. You are *not* adding new dynamical degrees of freedom; you are interrogating the ones you have. The current j^μ is the linear response: $\langle j^\mu(x) \rangle = -i \delta \log Z[A] / \delta A_\mu(x)$.

Example 6.1 (The simplest background gauge field: a particle on a ring)

Nothing illustrates “background field = knob” better than a free particle of unit charge on a circle of circumference 2π , threaded by Aharonov–Bohm flux θ . The flux is a flat background U(1) gauge field $A = \frac{\theta}{2\pi} dx$: locally pure gauge ($F = 0$), but with physical holonomy $\oint A = \theta$. Minimal coupling shifts the momentum, giving the spectrum

$$E_n = \frac{1}{2} \left(n - \frac{\theta}{2\pi} \right)^2, \quad n \in \mathbb{Z}.$$

Three lessons in one formula: (i) the spectrum depends *only* on the holonomy θ , not on the pointwise A —the background’s physical content is its holonomy; (ii) $\theta \sim \theta + 2\pi$ (relabel n), so the knob is U(1)-valued, as a flat U(1) background should be; (iii) at $\theta = \pi$ two levels cross—the first hint of the kind of degeneracy that a mixed ’t Hooft anomaly at $\theta = \pi$ enforces (§8). This toy model is the entire idea of $Z[A]$ stripped to one degree of freedom.^a

^aThis particle-on-a-ring / θ -angle picture, and the $\theta = \pi$ level crossing, are developed in Tong, *Lectures on Gauge Theory* §1.2, §2.2.3.

Equation 6.1 (Currents and Ward identities for free)

Differentiating $Z[A]$ in the background generates correlators of the current:

$$\langle j^{\mu_1}(x_1) \cdots j^{\mu_n}(x_n) \rangle = \frac{(-i)^n}{Z} \frac{\delta^n Z[A]}{\delta A_{\mu_1}(x_1) \cdots \delta A_{\mu_n}(x_n)} \Big|_{A=0}.$$

Background gauge invariance of $Z[A]$ is the statement that j^μ is conserved—the Ward identities are just $Z[A^\lambda] = Z[A]$ expanded in λ .

6.2 The defining property: background gauge invariance**Definition 6.2** (Having a symmetry, modern definition)

A theory **has** the global symmetry G (non-anomalously) if its partition function in a background can be made *invariant* under background gauge transformations,

$$Z[A^\lambda] = Z[A] \quad \text{for all } \lambda : M \rightarrow G,$$

possibly after adding local counterterms built from A . If no choice of local counterterms achieves

this—if $Z[A^\lambda] = Z[A] e^{i \int \omega(\lambda, A)}$ with a phase that cannot be removed—the symmetry has a **'t Hooft anomaly** (§8).

Intuition 6.2 (Why phrase symmetry this way?)

This definition is robust in ways the Noetherian “there is a conserved current” definition is not. It (1) works for finite symmetries with no current; (2) makes anomalies visible as a sharp obstruction (a residual phase); (3) is manifestly RG-invariant—if you can couple to a background in the UV you can in the IR, which is the engine of 't Hooft anomaly matching; (4) generalizes verbatim to higher-form symmetries by letting A be a higher-form background. This is why “couple to a background and study $Z[A]$ ” is the modern definition of a symmetry.

Example 6.2 (Free Dirac fermion in a background $U(1)$)

Take a free massless Dirac fermion in 4d with its vector $U(1)_V$ symmetry. Coupling to a background A_μ via $\not{D} = \gamma^\mu(\partial_\mu - iA_\mu)$ gives $Z[A]$. The vector current is conserved for all A : $Z[A]$ is $U(1)_V$ background-gauge-invariant. But the *axial* current j_5^μ is *not*: under an axial background gauge transformation Z picks up the famous phase $\propto \int \lambda F \wedge F$. That is the chiral (ABJ) anomaly, our first example of an 't Hooft anomaly, and it is why $\pi^0 \rightarrow \gamma\gamma$.

Intuition 6.3 (Why the chiral anomaly is unavoidable: spectral flow)

You can *see* the chiral anomaly without a single Feynman diagram. Put a 2d massless Dirac fermion in a box and fill the Dirac sea; left-movers and right-movers have separate Fermi seas. Now adiabatically turn on a background electric field $E = \partial_t A_x$ (a time-dependent holonomy of A). Each single-particle level's momentum shifts as $\dot{p} = E$, so states flow *up* through the right-moving sea and *down* through the left-moving sea: the number of right-movers and left-movers each change, even though total charge is conserved. The axial charge $Q_5 = N_R - N_L$ is therefore *not* conserved, at a rate fixed purely by how many levels crossed—i.e. by $\int E$. That counting is rigid (an integer flow), which is exactly why the anomaly coefficient cannot be tuned away by any local counterterm.^a This is the microscopic face of “the anomaly is a quantized obstruction.”

^aThe spectral-flow / level-crossing derivation is in Tong, *Lectures on Gauge Theory* §3.1; the relation to the Atiyah–Singer index and instanton zero modes is §3.3.

Example 6.3 (Background metric is a background gauge field for spacetime symmetry)

The stress tensor $T^{\mu\nu}$ is the current for translations; coupling it to a background means putting the theory on a curved metric $g_{\mu\nu}$, giving $Z[g]$. “Background diffeomorphism + Weyl invariance of $Z[g]$ ” is exactly the statement of no gravitational/conformal anomaly; the failure is the trace anomaly $\langle T^\mu_\mu \rangle \sim c(\text{curvature})$. *Same machine, symmetry = spacetime symmetry, background = metric*. Keep this analogy close: it tells you the metric is “just” the background gauge field

for the Poincaré/conformal symmetry.

Exercise 6.1 ((medium) current as response, explicitly) — For the complex scalar of §2 coupled to background A_μ , compute $\delta \log Z[A]/\delta A_\mu$ at $A = 0$ and check it equals $\langle j^\mu \rangle$ with $j^\mu = i(\phi^* \partial^\mu \phi - \phi \partial^\mu \phi^*)$. Then show invariance of $Z[A]$ under $A \rightarrow A + d\lambda$ at linear order reproduces $\partial_\mu j^\mu = 0$.

6.3 Backgrounds for finite symmetries: defect networks

For a finite symmetry G there is no current to write $\int A_\mu j^\mu$. Instead, turning on a background $a \in H^1(M; G)$ means *inserting a network of symmetry defects* Poincaré-dual to a .

Intuition 6.4 (Background = symmetry-defect network = twisted boundary conditions)

A G symmetry operator U_g is supported on a codimension-1 surface; crossing it acts on operators by g (§9). A background a specifies, for each codimension-1 sheet in a chosen dual cell decomposition, which U_g sits there. On a spacetime cycle, this is exactly a **twisted boundary condition**: “identify fields up to the action of g when you go around.” For $\mathbb{Z}_2$ on a circle, $a =$ nontrivial means antiperiodic/twisted; $Z[a]$ is the twisted partition function $\text{tr}(g e^{-\beta H})$.

Example 6.4 (Twisted partition functions of the Ising model)

The 2d Ising CFT has a $\mathbb{Z}_2$ spin-flip symmetry. Its torus partition functions in the four $\mathbb{Z}_2$ backgrounds $a \in H^1(T^2; \mathbb{Z}_2) = (\mathbb{Z}_2)^2$ are the four sectors $Z[a] = \text{tr}_{\mathcal{H}_b}(g^{a_t} e^{-\beta H})$ with periodic/twisted space (a_s) and time (a_t) directions. Modular transformations permute these four $Z[a]$ ’s; demanding consistency under gauging $\mathbb{Z}_2$ (summing them, §7) reproduces Kramers–Wannier duality. *This little 2×2 table of twisted partition functions is the entire input to a huge amount of modern 2d physics.*

6.4 Background vs. dynamical: the clean contrast

	Background gauge field	Dynamical gauge field
Role in path integral	fixed external data; <i>not</i> integrated over	a field; <i>integrated</i> over ($\int \mathcal{D}A$ or $\sum_a$)
Associated symmetry	a <i>global</i> symmetry it probes	gauge “symmetry” is a <i>redundancy</i> , not a symmetry
Equations of motion	none (it is a source)	has its own EOM / Gauss-law constraint
What you compute	$Z[A]$ as a functional; correlators of j	a new theory’s Z ; gauge-invariant operators only
Gauge transformations	a true symmetry of $Z[A]$ (or anomalous)	a redundancy you quotient by
Examples	external EM field, curved metric, $\mathbb{Z}_2$ twist μ	photon in QED, gluon, $\mathbb{Z}_2$ gauge theory

Note 6.1 (The slogan). **Background:** you fix it and study the response. **Dynamical:** you sum over it and get a new theory. The very same connection A can be either, depending on whether it appears under the path-integral measure. “Gauging” is the operation that turns the first into the second.

Exercise 6.2 ((easy) which is which?) — Classify each as background or dynamical, and name the (global or gauge) group: (a) the photon in QED; (b) the external magnetic field in the quantum Hall effect; (c) the gluon field; (d) the metric in a non-gravitational QFT; (e) the $\mathbb{Z}_2$ field recording fermion boundary conditions; (f) the $\mathbb{Z}_2$ field in the toric code.

7 Gauging: making a symmetry local

Definition 7.1 (Gauging)

To **gauge** a global symmetry G is to promote its background gauge field from fixed to dynamical: sum/integrate over A in the path integral. Schematically,

$$Z_{\text{gauged}} = \int \mathcal{D}A Z[A] \quad (\text{continuous}), \quad Z_{\text{gauged}}[M] = \frac{1}{|G|^{N_0}} \sum_{\{g_e\}} Z[\{g_e\}] \quad (\text{finite}).$$

For continuous G one also adds a dynamical kinetic term $\frac{1}{2g^2} \int \text{tr} F^2$. For finite G the sum runs over *all* link configurations $\{g_e\}$ and is divided by the number of gauge transformations $|G|^{N_0}$ (N_0 = number of vertices)—the volume of the gauge group, i.e. exactly one factor of $1/|G|$ per redundancy. Restricted to *flat* fields this collapses to a sum over $\text{Hom}(\pi_1(M), G)/\text{conj.}$ (equivalently $H^1(M; G)$ for abelian G , the configurations of §5). The result is a *new theory* in

which G is a redundancy.^a

^aPrecise lattice prescription (and its higher-form version): Kapustin–Seiberg 1401.0740 §8. Equivalently one writes the gauged partition function as a weighted sum over twisted sectors $Z_{\text{gauged}} = \sum_a h_a Z_a$ (GKSW 1412.5148, eq. 2.5); the freedom in the weights h_a is the discrete theta angle / discrete torsion of the next exercise.

Intuition 7.1 (Gauging in one breath)

“Gauging” literally means “declaring the symmetry to be a local redundancy of description rather than a physical operation.” Operationally you do this by making its gauge field dynamical. Before: A is a knob, the symmetry acts on states. After: A fluctuates, only G -invariant data is physical, and the symmetry no longer maps distinct physical states to one another—it relates descriptions of the *same* state. You can only do this if there is no ’t Hooft anomaly (§8): an anomaly is exactly the obstruction to making $Z[A]$ invariant, hence to summing over A consistently.

7.1 What gauging produces: new operators, new symmetries

Gauging changes the operator content in two systematic ways:

- **Projection.** Only G -invariant (gauge-invariant) operators survive. Charged local operators are projected out (or must be attached to Wilson lines).
- **New extended operators.** Wilson lines $\text{tr } \mathcal{P} \exp i \oint A$ and (for θ -like data) ’t Hooft/vortex operators appear, charged under *new* symmetries that did not exist before gauging—the **quantum** or **dual** symmetry. For gauging a finite abelian G in d dimensions you generate a dual $\widehat{G} = \text{Hom}(G, \text{U}(1))$ symmetry that is a $(d - 2)$ -form symmetry (§9).

Example 7.1 (Gauging $\text{U}(1)$: scalar QED, and a dual current)

Gauging the global $\text{U}(1)$ of the complex scalar (§2) by making A_μ dynamical gives scalar QED. The would-be charged operator ϕ is no longer gauge invariant on its own. In 3d a new $\text{U}(1)_{\text{top}}$ “topological” symmetry appears with current $j_{\text{top}}^\mu = \frac{1}{2\pi} \epsilon^{\mu\nu\rho} \partial_\nu A_\rho$, whose charge is the magnetic flux; in 4d the analogous structure is the magnetic 1-form symmetry with current $\star F$. *Gauging gives with one hand (a dual symmetry) what it takes with the other (the original charged operators).*

Example 7.2 (Gauging $\mathbb{Z}_2$ in 2d: orbifolds and Kramers–Wannier)

Gauge the $\mathbb{Z}_2$ symmetry of the 2d Ising model by summing $Z[a]$ over the four backgrounds $a \in H^1(T^2; \mathbb{Z}_2)$ with the right normalization. The result is again an Ising model—but with a *dual* $\widehat{\mathbb{Z}_2}$ symmetry exchanged with the original. This is Kramers–Wannier duality realized as “gauging a $\mathbb{Z}_2$.” In string theory the very same sum over $Z[a]$ is an **orbifold**. Gauging a finite symmetry = orbifolding; the dual symmetry $\widehat{\mathbb{Z}_2}$ is the “quantum symmetry” of the orbifold. *This single example—gauge $\mathbb{Z}_2$, get dual $\mathbb{Z}_2$, recover a duality—is the prototype for an enormous*

amount of modern QFT.

Exercise 7.1 ((medium) gauging twice returns home) — Argue that gauging a finite abelian G produces a theory with dual symmetry $\widehat{G}$, and that gauging $\widehat{G}$ in turn recovers the original theory. (In 2d, $\widehat{G} \cong G$, so “gauge $\mathbb{Z}_2$ twice = identity.”) *This is the abstract content of Kramers–Wannier self-duality.*

Exercise 7.2 ((thinker) discrete theta angle / choice of gauging) — When you gauge G you may add a phase weight to each background: a **discrete theta angle** valued in $H^d(BG; \mathbb{U}(1))$ ($\sim$ a Dijkgraaf–Witten term). For $G = \mathbb{Z}_2$ in 2d, $H^2(B\mathbb{Z}_2; \mathbb{U}(1)) = 0$, but in 3d, $H^3(B\mathbb{Z}_2; \mathbb{U}(1)) = \mathbb{Z}_2$ gives *two* inequivalent ways to gauge. Explain why “how you gauge” is extra data, not canonical, and connect it to stacking with an SPT before gauging.

7.2 Gauging a spacetime symmetry; subtleties

Example 7.3 (Gauging gives gravity / orientifolds / time-reversal subtleties)

Gauging translations & rotations (making the metric dynamical) gives gravity. Gauging a $\mathbb{Z}_2$ reflection or time-reversal gives unoriented/pin geometries (orientifolds). These cases carry extra subtleties (you must sum over geometries, spin structures, orientations), but the logic is identical: take a background (the metric, the orientation), make it dynamical.

Note 7.1 (Gauging is not free: the anomaly gate). You cannot gauge a symmetry with an 't Hooft anomaly: the sum $\sum_a Z[a]$ (or $\int \mathcal{D}A Z[A]$) is not well defined / not gauge invariant precisely because $Z[A^\lambda] \neq Z[A]$. The anomaly is the gate that decides whether gauging is even possible. That is why anomalies, which sound like a technical pathology, are actually the central organizing constraint of the modern program—next section.

8 't Hooft anomalies and anomaly inflow

8.1 Definition and the meaning of the phase

Definition 8.1 ('t Hooft anomaly)

A global symmetry G has an **'t Hooft anomaly** if, after coupling to a background A , the partition function cannot be made background-gauge-invariant by any local counterterm; instead

$$Z[A^\lambda] = Z[A] \exp\left(i \int_{M_d} \alpha(\lambda, A)\right),$$

with a definite, counterterm-independent phase α . Crucially, $Z[A]$ is still well defined and the symmetry still acts on the Hilbert space (this is *not* the inconsistency of a gauged anomaly); what fails is only background gauge invariance.

Intuition 8.1 (Anomaly = an obstruction you cannot localize away)

A would-be symmetry violated by a local, removable term is no anomaly—just a bad regulator. An 't Hooft anomaly is a violation that no local counterterm can absorb. The sharp test: can you write $Z[A]$ as the boundary value of a *local* $(d+1)$ -dim functional and restore invariance? If not, the leftover is the anomaly. This obstruction is quantized (it lives in a discrete classification), hence rigid: it cannot drift under continuous deformations, which is what makes it RG-invariant.

8.2 Anomaly inflow: the bulk SPT**Equation 8.1** (The inflow picture)

An anomalous d -dimensional theory is the boundary of a $(d+1)$ -dimensional **invertible**/SPT theory:

$$Z_d[A] = (\text{boundary of}) \exp\left(i \int_{X_{d+1}} \omega(A)\right), \quad \partial X_{d+1} = M_d.$$

The bulk term $\omega(A)$ is gauge invariant in the closed bulk; its variation localizes to the boundary and produces exactly the anomalous phase α . The anomaly is *classified* by which bulk SPT you need: an element of group cohomology $H^{d+1}(BG; \mathbb{U}(1))$ (or, more completely, of a cobordism group Ω_G^{d+1}).^a

^aAnomalies as boundary modes of invertible/SPT phases, and their cobordism classification, are developed in Tachikawa's TASI 2019 lectures *Anomalies and topological phases* and his *Lectures on topological phases for mathematicians*.

Example 8.1 (The 2d chiral fermion's anomaly inflow)

A single right-moving complex fermion in 2d has a $\mathbb{U}(1)$ anomaly; it lives on the boundary of a 3d $\mathbb{U}(1)$ Chern–Simons term $\frac{k}{4\pi} \int A \, dA$. The non-invariance of CS on a manifold with boundary is precisely the 2d anomalous phase. This is the field-theoretic skeleton of the integer quantum Hall edge: a $(2+1)$ d bulk with a chiral edge that cannot exist on its own.

Example 8.2 (Mixed anomaly: momentum vs. winding of the 2d compact boson)

The free compact boson $\phi \sim \phi + 2\pi$ in 2d has two $\mathbb{U}(1)$ symmetries: a **momentum** symmetry ($\phi \rightarrow \phi + c$, current $\propto \star d\phi$) and a **winding** symmetry (current $\frac{1}{2\pi} d\phi$, charge = winding number). Couple backgrounds A_m, A_w for both. No local counterterm makes $Z[A_m, A_w]$ invariant under both at once; the leftover is a **mixed anomaly** with bulk inflow

$$Z[A_m^\lambda, A_w] = Z[A_m, A_w] \exp\left(\frac{i}{2\pi} \int_{M_2} \lambda \, dA_w\right), \quad \text{3d SPT } \frac{1}{2\pi} \int_{X_3} A_m \wedge dA_w.$$

(Note the bulk term is a genuine 3-form integrated over a 3-manifold—contrast a mass-dimension slip like “ $\int_{X_3} a_1 \cup a_2$,” which is only a 2-form.) **Consequence:** you cannot gauge momentum and winding simultaneously. Gauging the momentum $\mathbb{U}(1)$ is T-duality, which exchanges the two symmetries (and inverts the radius); the anomaly is precisely why the theory cannot be trivially gapped while preserving both—consistent with the free boson being a CFT. *This is*

the cleanest mixed anomaly in physics, and it secretly powers a duality you already know.

8.3 't Hooft anomaly matching: the payoff

Theorem 8.1 ('t Hooft anomaly matching)

Because the anomaly is a discrete, deformation-invariant obstruction, it is the *same* in the UV and the IR of any RG flow. Therefore whatever degrees of freedom the theory flows to must reproduce the UV anomaly. In particular, a nonzero anomaly forbids a trivially gapped symmetric vacuum: the IR must contain massless particles, or spontaneously break the symmetry, or be a nontrivial TQFT.

Intuition 8.2 (Why this is the workhorse of hep-th)

Anomaly matching is one of the very few *exact, nonperturbative* constraints we have on strongly-coupled QFT.^a 't Hooft's original use: the anomalies of QCD's chiral $SU(N_f) \times SU(N_f)$ symmetry, computed on free quarks in the UV, must be matched in the confined IR—either by massless baryons (hard) or by chiral symmetry breaking with Goldstones (what nature does). Modern uses: ruling out trivial gapped phases, deriving the existence of edge modes, constraining the IR of $\theta = \pi$ gauge theories (Gaiotto–Kapustin–Komargodski–Seiberg), and constructing deconfined criticality.^b This is precisely the “Seiberg-style” reasoning you are heading toward.

^aThe matching principle originates in 't Hooft's 1980 Cargèse lecture *Naturalness, chiral symmetry, and spontaneous chiral symmetry breaking*.

^bThe $\theta = \pi$ mixed anomaly between time-reversal and the center one-form symmetry, and its IR consequences, are worked out in Gaiotto–Kapustin–Komargodski–Seiberg, *Theta, Time-Reversal, and Temperature*, [arXiv:1703.00501](#)—the template for end-of-chapter Exercise 10.6.

Exercise 8.1 ((classic) anomaly forbids a trivial gap) — Suppose a theory has a G symmetry with a nonzero 't Hooft anomaly. Argue that it *cannot* have a unique, symmetric, gapped ground state on every spatial manifold. *Hint: a trivially gapped symmetric phase has $Z[A] \rightarrow$ a manifestly invariant local counterterm in the deep IR, contradicting the anomalous phase.*

Exercise 8.2 ((thinker) anomaly vs. gauging, made precise) — Explain, using $Z[A^\lambda] = Z[A]e^{i \int \alpha}$, why an 't Hooft anomaly is exactly the obstruction to gauging G . Then explain why a *mixed* anomaly between G_1 and G_2 still allows gauging G_1 *alone*, but at the cost of the dual symmetry inheriting an anomaly / G_2 becoming anomalous or extended. *This is the “anomaly is conserved under gauging” principle.*

9 The modern point of view, and a preview of generalized symmetries

We can now assemble the pieces into the picture that organizes contemporary hep-th.

Equation 9.1 (The modern point of view, in one box)

A QFT is characterized by its **symmetries**; a symmetry is characterized by how it **couples to background gauge fields**, i.e. by the functional $Z[A]$ together with its (possibly anomalous) transformation $Z[A^\lambda] = Z[A]e^{i \int \alpha}$. From this data one reads off conserved currents (derivatives of Z), selection rules (Ward identities), 't Hooft anomalies (the phase α , classified by a bulk SPT), and the possible **gaugings** (sum over A , allowed iff $\alpha = 0$ for that subgroup), which generate **dualities** and **new (dual) symmetries**. Everything is phrased in the language of backgrounds and defects rather than Lagrangians.

9.1 Symmetries as topological operators**Idea 9.1** (The Gaiotto–Kapustin–Seiberg–Willett reframing)

A global symmetry is a **topological operator**. For an ordinary (0-form) symmetry G , the operator $U_g(\Sigma_{d-1})$ lives on a codimension-1 surface Σ , and “acting with the symmetry” is sweeping U_g across space. Topological means correlators are unchanged under continuous deformations of Σ that don’t cross charged insertions—which is just current conservation $d \star j = 0$ restated. A charged local operator $\mathcal{O}(x)$ is detected because moving U_g *through* x multiplies $\mathcal{O}$ by its charge: the symmetry operator and the charged operator [link](#). Background gauge field A = an inserted network of these U_g ’s.

Remark 9.1 (The unification: transition functions *are* the defect network). This closes a loop opened in §4. There, a G -bundle was specified by patches glued with transition functions g_{ij} on overlaps (satisfying the cocycle condition on triple overlaps). Here, a background flat connection is a network of defects U_g on codimension-1 sheets, joined at junctions (satisfying a consistency condition where sheets meet). *These are the same data*: put a defect $U_{g_{ij}}$ on the wall dual to each overlap $U_i \cap U_j$, and the cocycle condition $g_{ij}g_{jk} = g_{ik}$ becomes the requirement that the defects fuse consistently at the triple-overlap junction. So “ $\mathbb{Z}_2$ gauge field” (§5), “transition functions of a bundle” (§4), and “network of $\mathbb{Z}_2$ symmetry defects” (§6.3) are three names for one object. An ’t Hooft anomaly is then the statement that no consistent set of junction phases exists.^a

^aThis defect/transition-function dictionary, and anomalies as missing topological junctions, are GKSW 1412.5148 §2.

Intuition 9.1 (Why “topological operator” is the right notion)

This definition needs no Lagrangian, no Noether current, no continuity of G . It applies to finite symmetries (then U_g is the symmetry defect of §6.3), to spacetime symmetries, and—this is the leap—to symmetries whose charged objects are *extended*. Once a symmetry is “a topological operator that links charged objects,” nothing forces the charged object to be a point. That single relaxation generates the entire zoo of generalized symmetries.

9.2 Higher-form symmetries

Definition 9.1 (*p*-form symmetry)

A ***p*-form global symmetry** has symmetry operators $U_g(\Sigma_{d-1-p})$ supported on codimension- $(p+1)$ surfaces, acting on p -dimensional charged objects (e.g. Wilson lines for $p = 1$). It is probed by a ***(p+1)*-form background gauge field** $B^{(p+1)}$. Ordinary symmetries are the $p = 0$ case. The group of a p -form symmetry is always abelian for $p \geq 1$.

Example 9.1 (Free Maxwell theory has two 1-form symmetries)

Source-free Maxwell in 4d obeys $dF = 0$ and $d \star F = 0$. Each closed 2-form is a conserved current for a 1-form symmetry:

$$\text{electric } U(1)_e^{(1)} : \oint_{S^2} \star F, \quad \text{magnetic } U(1)_m^{(1)} : \frac{1}{2\pi} \oint_{S^2} F.$$

The charged objects are Wilson and 't Hooft lines respectively. There is a *mixed* 't Hooft anomaly between them (you cannot gauge both), encoded by the 5d term $\int B_e dB_m$. This reframes electromagnetic duality as the statement that these two 1-form symmetries are exchanged. *Maxwell theory, the thing you learned first, is the cleanest laboratory for the newest ideas.*

Example 9.2 (Center symmetry and confinement in pure Yang–Mills)

Pure $SU(N)$ Yang–Mills has a $\mathbb{Z}_N^{(1)}$ **center** 1-form symmetry acting on Wilson lines. Its order parameter is the Wilson loop: an area law $\langle W(C) \rangle \sim e^{-(\text{area})}$ signals *unbroken* $\mathbb{Z}_N^{(1)}$, i.e. **confinement**; a perimeter law $e^{-(\text{perimeter})}$ signals *broken* $\mathbb{Z}_N^{(1)}$, i.e. deconfinement. Confinement, the hardest qualitative fact about QCD, becomes the statement “the 1-form center symmetry is unbroken.” Adding dynamical quarks in the fundamental explicitly breaks $\mathbb{Z}_N^{(1)}$ (they can end Wilson lines), which is why the order parameter is only sharp in the pure-gluon theory. *This is the single example that most convinced people higher-form symmetries are physics, not formalism.*

Exercise 9.1 ((medium) higher-form gauge fields are higher cochains) — Convince yourself that a $\mathbb{Z}_N$ 1-form symmetry’s background is a 2-cochain $B \in C^2(M; \mathbb{Z}_N)$ with gauge transformation $B \rightarrow B + d\lambda^{(1)}$, exactly mirroring §5. State the analogue of “flat” and of H^1 for this case. *This is why the $\mathbb{Z}_2$ 2-form gauge field B you see in papers is “just” a 2-cochain.*

Exercise 9.2 ((thinker) count the symmetry structure of $U(1)$ Maxwell on T^4) — Using $H^2(T^4; U(1))$, count the sectors/fluxes that the electric and magnetic 1-form symmetries organize, and identify the order parameters (Wilson and 't Hooft surfaces wrapping 2-cycles). *This is a finger exercise for the kind of bookkeeping generalized-symmetry papers do constantly.*

9.3 The bridge to Seiberg-style QFT

Equation 9.2 (The core dictionary, and the “five moves”)

Keep this one table and these five verbs and you can read most of the modern literature.

object	is
G -gauge field	a connection on a principal G -bundle (finite G : a G -cochain)
curvature F	holonomy per unit area; $F = 0 \Leftrightarrow$ transport is topological
background field A	a fixed probe; the data is the functional $Z[A]$
symmetry	a topological operator that links its charged objects
't Hooft anomaly	the phase $Z[A^\lambda]/Z[A]$; a bulk SPT; obstructs gauging

The five moves: (1) *couple* a symmetry to a background A ; (2) read off currents/Ward identities as derivatives of $Z[A]$; (3) detect the *anomaly* as the non-removable phase of $Z[A^\lambda]$; (4) *gauge* (sum over A) when the anomaly vanishes, producing a dual symmetry and often a duality; (5) *match* anomalies across RG to constrain the IR.

Preview 9.1 (Where to go next)

You now have the working vocabulary for current hep-th: gauge fields are connections; backgrounds probe symmetries through $Z[A]$; gauging sums over backgrounds and produces dual symmetries and dualities; 't Hooft anomalies (classified by bulk SPTs / cobordism) obstruct gauging and constrain RG flows; and “symmetry” has been generalized to topological defect operators acting on extended objects (p -form, and beyond: non-invertible symmetries, 2-groups, symmetry TFTs). The natural next steps are (i) generalized symmetries in detail (GKSW; the review literature in the references), (ii) the SymTFT (“symmetry topological field theory”) packaging of $Z[A]$ and its anomaly as a bulk theory, and (iii) the use of all this to constrain and solve strongly-coupled theories (Seiberg dualities, $\mathcal{N}=1, 2, 4$ gauge theories, deconfined criticality). Everything there is built on the five moves in these notes.

10 End-of-chapter exercises

These are meant to be worked after a full read. Difficulty tags as before. A few are open-ended “thinkers” designed to push you toward the literature.

Exercise 10.1 ((classic) the gauging algorithm by hand) — Start from a free massless Dirac fermion in 4d with global $U(1)_V$. (a) Couple to a background A_μ and write $Z[A]$. (b) Verify $U(1)_V$ background gauge invariance. (c) Now make A_μ dynamical (add a Maxwell term) to get QED; list which operators are gauge invariant and identify the magnetic 1-form symmetry. (d) State precisely what would have gone wrong had you tried to gauge the *axial* $U(1)_A$ instead.

Exercise 10.2 ((medium) monopole bundle arithmetic) — For a charge- n Dirac monopole, construct A_N, A_S on the two hemispheres explicitly, find the transition function on the equator,

and verify $\frac{1}{2\pi} \int_{S^2} F = n$. Then show a charge- q field is consistent only if $qn \in \mathbb{Z}$ (Dirac–Schwinger–Zwanziger in its simplest form).

Exercise 10.3 ((easy) reading comprehension on conventions) — A paper writes “let a be a $\mathbb{Z}_2$ one-form gauge field and b a $\mathbb{Z}_2$ two-form gauge field, with action $\frac{2\pi}{2} \int a \cup b$.” In your own words: what kinds of symmetries are being backgrounded, what cohomology groups do a, b live in, and what does the cup-product action encode?

Exercise 10.4 ((medium) the toric code as gauged $\mathbb{Z}_2$) — Take the 2d quantum Ising model and gauge its $\mathbb{Z}_2$ symmetry on the lattice (Wegner’s construction). Identify the resulting model as the toric code, find the two types of anyon (electric charge e and flux m), and explain how e comes from the projected Ising spin and m from the dual/quantum $\widehat{\mathbb{Z}_2}$ symmetry. *This ties §5, §6, §7 together in a single concrete model.*

Exercise 10.5 ((classic) chiral anomaly as inflow) — Compute (or look up and then reproduce the logic of) the 2d abelian anomaly $\partial_\mu j_5^\mu \propto \epsilon^{\mu\nu} F_{\mu\nu}$, and show it is the boundary avatar of the 3d Chern–Simons term. State the 4d analogue $\partial_\mu j_5^\mu \propto F \wedge F$ and its 5d inflow.

Exercise 10.6 ((thinker) $\theta = \pi$ and a mixed anomaly) — In 4d $U(1)$ (or $SU(2)$) gauge theory at $\theta = \pi$, time-reversal T is a symmetry. Argue there is a mixed ’t Hooft anomaly between T and the 1-form (or center) symmetry, and state what it forbids for the IR (following Gaiotto–Kapustin–Komargodski–Seiberg). *This is a genuine modern research-level result; aim to understand the statement and the strategy, not necessarily to rederive every step.*

Exercise 10.7 ((thinker) non-invertible symmetry teaser) — Gauging a $\mathbb{Z}_2$ subgroup of a larger symmetry, or stacking duality defects, can produce a symmetry whose defects do *not* have inverses (the fusion of defects gives a sum, not a single defect). Read the statement of Kramers–Wannier duality as a *non-invertible* symmetry of the Ising CFT at criticality, and explain in words why “gauging on half of spacetime” produces a non-invertible defect. *This is the current frontier; the references point the way.*

Exercise 10.8 ((medium) build your own dictionary) — Without looking back, fill in this table from memory, then check: for each of $U(1)$ 0-form, $\mathbb{Z}_2$ 0-form, $U(1)$ 1-form, $\mathbb{Z}_N$ 1-form—give (i) the background gauge field (form degree and value group), (ii) the charged objects, (iii) the symmetry operators’ codimension, (iv) one physical example.

References and a reading plan

I wrote these notes from the following sources; they are also, in order, a sensible reading plan. I have flagged which are best for the *geometry* (Parts I–II) and which for the *modern POV* (Part III).

Lecture notes and papers (freely available; good to load into memory).

- **D. Tong**, *Lectures on Gauge Theory* and *Lectures on Quantum Field Theory* (damtp.cam.ac.uk/user/tong/teaching.html). The friendliest modern entry to Parts I and III; superb on instantons, θ , and anomalies.
- **D. Harlow**, *8.325: Quantum Field Theory III* lecture notes (MIT). The cleanest physicist's account of the Lie-algebra-valued connection (§3, the basis for the expanded §3.1 here) and of Hamiltonian lattice gauge theory (§8, “what is a gauge field?”).
- **Y. Tachikawa**, TASI 2019 lectures (topological phases and anomalies) and his lecture notes on symmetries and gauging (member.ipmu.jp/yuji.tachikawa/lectures). The cleanest treatment of “ $Z[A]$, gauging finite symmetries, anomalies as bulk SPTs.”
- **Gaiotto, Kapustin, Seiberg, Willett**, *Generalized Global Symmetries*, JHEP (2015), arXiv:1412.5148. The founding paper of Part III; read §2 carefully.
- **Kapustin, Seiberg**, *Coupling a QFT to a TQFT and Duality*, arXiv:1401.0740. The precise mechanics of background/dynamical finite gauge fields and discrete theta angles.
- **Gaiotto, Kapustin, Komargodski, Seiberg**, *Theta, Time-Reversal, and Temperature*, arXiv:1703.00501. The flagship “anomaly $\Rightarrow$ nontrivial IR” application; the model for end-of-chapter Exercise on $\theta = \pi$.
- **J. McGreevy**, *Generalized Symmetries in Condensed Matter*, Ann. Rev. Cond. Mat. (2023), arXiv:2204.03045. Outstanding on the lattice/ $\mathbb{Z}_2$ side and on why this is physics.
- **Córdova, Dumitrescu, Intriligator, Shao**, *Snowmass White Paper: Generalized Symmetries*, arXiv:2205.09545; the review **Schafer-Nameki et al.**, *Lectures on Generalized Symmetries*, arXiv:2307.07547; and **Sharpe**, *Notes on generalized global symmetries*, arXiv:1508.04770. For the road past these notes.
- **'t Hooft** (1980), *Naturalness, chiral symmetry, and spontaneous chiral symmetry breaking*—the origin of anomaly matching. **Wegner** (1971) and **Kogut**, *Rev. Mod. Phys.* **51** (1979)—the origin of $\mathbb{Z}_2$ /lattice gauge theory. **Dijkgraaf–Witten** (1990)—finite gauge theory and discrete theta.
- **Wu–Yang** (1975), *Concept of nonintegrable phase factors and global formulation of gauge fields*—the paper that put bundles into physics (§4).

Books (the ones I would buy/borrow—tell me which to load).

- **M. Nakahara**, *Geometry, Topology and Physics*. The standard physicist's reference for bundles, connections, characteristic classes (Parts I–II).
- **J. Baez & J. Muniain**, *Gauge Fields, Knots and Gravity*. The most *intuitive* book on the connection/holonomy picture; read alongside §2–§4.
- **M. Hamilton**, *Mathematical Gauge Theory*. Modern, rigorous, and physics-motivated; the best single book for the precise bundle statements in §4 (does the Standard Model carefully).
- **T. Frankel**, *The Geometry of Physics*. Broad and excellent for forms, bundles, and the $\text{GR} \leftrightarrow \text{YM}$ dictionary.

- **G. Naber**, *Topology, Geometry and Gauge Fields* (2 vols). For going deep on instantons, monopoles, and the topology.
- **Kobayashi & Nomizu**, *Foundations of Differential Geometry*. The pure-math source for connections on principal bundles, if you want full rigor.
- **E. Fradkin**, *Quantum Field Theory: An Integrated Approach* (and *Field Theories of Condensed Matter Physics*). Excellent on lattice/ $\mathbb{Z}_2$ gauge theory, the toric code, and gauging (Parts II–III).
- **Peskin & Schroeder** (Ch. 15, 16, 19) and **Weinberg** (Vol. II) for the standard QFT treatment of Yang–Mills and the chiral anomaly. **R. Bertlmann**, *Anomalies in Quantum Field Theory*, for everything anomaly with the geometry made explicit (§8).

Suggested order. Baez–Muniain or Tong (Parts I–II) $\rightarrow$ Nakahara or Hamilton for rigor $\rightarrow$ Tachikawa’s notes + GKS §2 (Part III) $\rightarrow$ McGreevy for the condensed-matter/lattice perspective $\rightarrow$ GKKS 1703.00501 for a full-strength application $\rightarrow$ the generalized-symmetries reviews for the frontier.